

HIRAM ISRAEL MENDOZA LOPEZ

COMPUTER SCIENCE & TECHNOLOGY GRADUATE | SOFTWARE DEVELOPER

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HIGHLIGHTED PROJECTS

Plagiarism Detection System – Siamese Bi-LSTM Neural Network

Queretaro, Mexico

Machine Learning Developer | Python, TensorFlow/Keras, NLP

Feb 2025 - Jun 2025

Academic project: deep learning system to detect plagiarism between Python code submissions, trained on a 20,000-pair dataset (PoolC + custom augmentation), balanced 50/50 between clones and non-clones.

- Designed and implemented the stratified train/test split strategy, ensuring balanced representation across clone types and establishing a reliable evaluation foundation for the team's models.
- Developed the V1 concatenation model, an early Siamese Bi-LSTM architecture for encoding and comparing Python code pairs, which served as the baseline for the team's subsequent iterations.
- Contributed to a final model (V2) that reached 84.55% accuracy (F1 0.846), with near-perfect recall (~99-100%) on surface-level plagiarism (Types 1-3) but notably lower recall (69.8%) on semantic plagiarism (Type 4) .

Altextex – Modular Enterprise System

Queretaro, Mexico

Software Architect & Full Stack Developer | React, Node.js, Express.js, MySQL, AWS S3, JWT, Swagger

Feb 2025 - Jun 2025

Enterprise project developed under a quality-focused approach aligned with CMMI and Disciplined Agile Delivery (DAD).

- Designed and implemented a modular architecture using Hexagonal Architecture and Clean Architecture principles.
- Developed enterprise modules integrating Node.js backend services, MySQL databases, and a modular frontend architecture, contributing to 100% delivery of defined requirements.
- Applied CMMI-based quality practices, including technical documentation, code reviews, defect tracking, and functional testing.
- Led an 8-member team, coordinating planning, task distribution, partner communication, and strategic re-estimations during project deviations.

Queretaro Wushu Kung Fu Association

Queretaro, Mexico

Team Leader & Full Stack Developer | Swift, SwiftUI, iOS, MVVM

Sep 2024 - Dec 2024

iOS application for documentation, organization, and internal management of a Wushu academy.

- Led a 5-member team in the development of an iOS wiki and organization platform.
- Translated functional and non-functional requirements into an MVVM-based iOS architecture focused on scalability and maintainability.
- Implemented key SwiftUI features and conducted functional testing under Scrum methodology.

EDUCATION

Tecnológico de Monterrey (ITESM)

Queretaro, Mexico

B.S. in Computer Science and Technology

Aug 2022 - Jun 2026

Relevant Coursework: Data Structures, Algorithms, Databases, Software Engineering, Computer Networks, Cybersecurity.

University of Twente | Exchange Studies

Enschede, Netherlands

Exchange Semester in Computer Science

Aug 2025 - Feb 2026

Relevant Coursework: Software Development, Web Science, Serious Gaming.

SKILLS

- Languages:** Spanish native, English C1, German A2.
- Programming:** JavaScript, Python, SQL, Swift.
- Frontend/Mobile:** React, SwiftUI, Material UI, Bootstrap.
- Backend & Cloud:** Node.js, Express.js, REST APIs, MySQL, AWS S3.
- Architecture & Quality:** Hexagonal Architecture, Clean Architecture, MVVM, CMMI, IEEE 829.
- Tools & Methodologies:** Git, GitHub, Swagger, ESLint, Scrum, DAD, functional/integration testing.
- Soft Skills:** Leadership, stakeholder communication, problem-solving, adaptability, accountability.